Paderborn B01: preprocessing and analysis

What I did with experiment B01, what the data shows, and whether it can train the Machine Guardian multi-task model.

1 November 2026 · MAC Attack project · scripts `model/b01_preprocess.py`, `model/b01_analysis.py` · outputs `model/results/B01/` · generated by `model/report/make_b01_report.py`

377 / 377	**3 segments**	**M0208**	**No**
measurements read, all aligned with the CSV records	two stops; 272 records after the log's 09:13 end	abrupt vibration change, 10:06:24, not explained by speed, load or temperature	reliable labels or 0–100% health index from this run alone

> **Bottom line.** B01 is well recorded and complete, and it shows one clear vibration change. It is one bearing, with no documented condition, and most of it is not covered by the experiment log. It is a good test case for change detection, but not enough to train or validate the deterioration and health-index outputs: more Paderborn experiments and documentation are needed first (section 8).

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1. What I was asked and what I did

Goal: prepare B01 for the multi-task model (vibration, temperature and RPM → deterioration detection and a 0–100% Motor Health Index), analysing only; no network was trained or changed, and the Ottawa model and files were not touched.

1. **Located and inspected the dataset** in `model/newdata/B01/`: 377 `.mat` files, two CSV files and a 2-page experiment log (read in the PDF viewer, because part of its text could not be extracted).
2. **Read every vibration file** and computed, for both accelerometers (rear A, front C), converted from volts to g with 10 g/V: mean, standard deviation, RMS, peak, peak-to-peak, crest factor, skewness, kurtosis, and RMS in four frequency bands (10 Hz–1 kHz, 1–5 kHz, 5–20 kHz, 20–64 kHz).
3. **Merged** them with T1, T2, room temperature, set and measured speed, static and dynamic load and timestamps into one chronological table, `B01_processed.csv` (377 rows, 47 columns).
4. **Checked integrity:** missing files, sample counts, timestamps, the file-to-record alignment, interruptions, and every operating condition against the log's ranges (`checks.json`).
5. **Investigated the log/CSV discrepancy** (the log ends at 09:13, the data at 10:59).
6. **Plotted** vibration, temperature, speed and load over time, plus vibration features (5 figures).
7. **Looked for deterioration onset** with speed and load removed, and tested whether warm-up, restarts or temperature could explain what was found (`analysis.json`).
8. **Wrote** `B01_analysis_report.md` and this PDF, and added `model/newdata/` to `.gitignore` so the 1.1 GB dataset cannot be committed.

Nothing was invented: no missing value was filled, and no deterioration label, failure time or health percentage was written to any file.

2. The dataset

Item	Content
Vibration	377 files `data_B01_M0001.mat` – `M0377`; two horizontal accelerometers (rear A, front C), 204,800 samples at 128 kHz = 1.6 s, in volts (10 g/V); about 12 s pause between measurements
Temperatures	`B01_meanTemperatures.csv`: bearing temperatures T1, T2 and room temperature, one row per measurement
Operating conditions	`B01_operatingConditions.csv`: set and measured speed (rpm), set and measured static load, set and peak dynamic load (N)
Protocol (log)	new bearing (run at most 15 min the day before); warm-up 10 min at 750 rpm, 2500 N, no dynamic load; then random speed 2400–3000 rpm, static 3300–3800 N, dynamic amplitude 500–1400 N at 2 Hz, combined 4300–4900 N
Log entries	19 Oct 2021, 08:45 start; 09:13 "the experiment successfully completes because the temperature threshold value was exceeded"
Missing link	the `.mat` files have no clock time and the CSVs no measurement id

3. Integrity checks

Check	Result
Files present	377 of 377, ids contiguous, none missing or extra
Signal format	204,800 samples at 128,000 Hz in every file; no missing or non-finite values
CSV timestamps	identical in both files, strictly increasing, 0 duplicates; 11–26 s between records in operation (median 18 s)
Alignment test 1	the warm-up ends after row 51; vibration (log RMS) jumps by 2.00 between M0051 and M0052, against at most 0.045 within the warm-up
Alignment test 2	step-to-step vibration changes follow speed changes at zero offset ($r = +0.24$) and at no other offset ($\lvert r\rvert \le 0.11$ for offsets −3 to +3)
Conclusion	measurement k belongs to CSV row k

4. The log ends at 09:13, the data at 10:59

The data do not describe one uninterrupted run. They split into three segments at two stops:

Segment	Measurements	Time	T1 start → end	Notes
1	M0001–M0105 (105)	08:46:54–09:13:24	21.4 → 113.6 °C	includes the 51-measurement warm-up; ends where the log says the temperature threshold was exceeded
stop		09:13:24–09:32:31	113.6 → 51.1 °C	19 min 7 s: the bearing cooled down
2	M0106–M0133 (28)	09:32:31–09:40:50	51.1 → 89.1 °C	restarted at full operating conditions, **without warm-up**
stop		09:40:50–09:43:08	89.1 → 82.1 °C	2 min 18 s
3	M0134–M0377 (244)	09:43:08–10:59:32	82.1 → 119.0 °C (max 122.6)	not in the log; T1 stays above segment 1's stopping temperature for over an hour; end reason not recorded

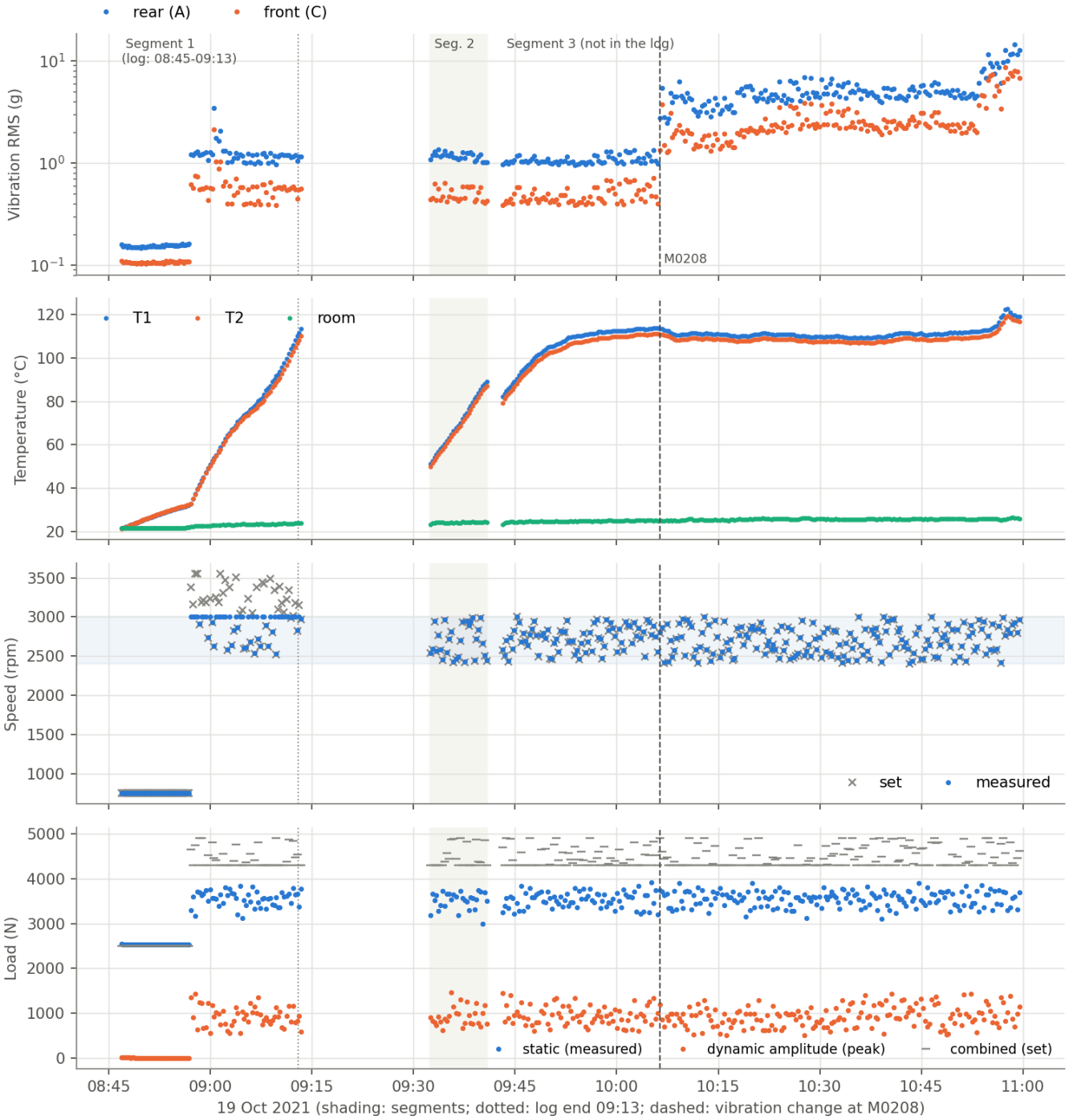
Segments 2 and 3 continue the same file numbering, acquisition and random protocol, and their vibration at the restart matches the end of segment 1, so they appear to be the same bearing. The threshold was probably raised or disabled, but none of this is documented: 272 of 377 measurements (72%) are outside the log. B01 is treated as three segments of one bearing with unknown interventions between them.

Operating conditions against the log

- **Speed:** 33 setpoints, all in segment 1, are above the logged 2400–3000 rpm (up to 3552 rpm); in 32 of them the measured speed stopped at about 3003 rpm. Use the measured speed, not the setpoint.

- **Combined load:** always within 4300–4900 N, but exactly 4300 N in 172 of 326 operating rows (the random draws were clipped to the range limit).
- **Static load:** measured minus set is 70 N in the median; 48 rows are off by more than 150 N (up to 552 N).

5. Sensor measurements over time



Vibration RMS of both accelerometers (log scale), temperatures, set and measured speed (shaded band: the logged 2400–3000 rpm), and loads. Shading marks the segments; the dotted line is the log's 09:13 end; the dashed line is the vibration change at M0208.

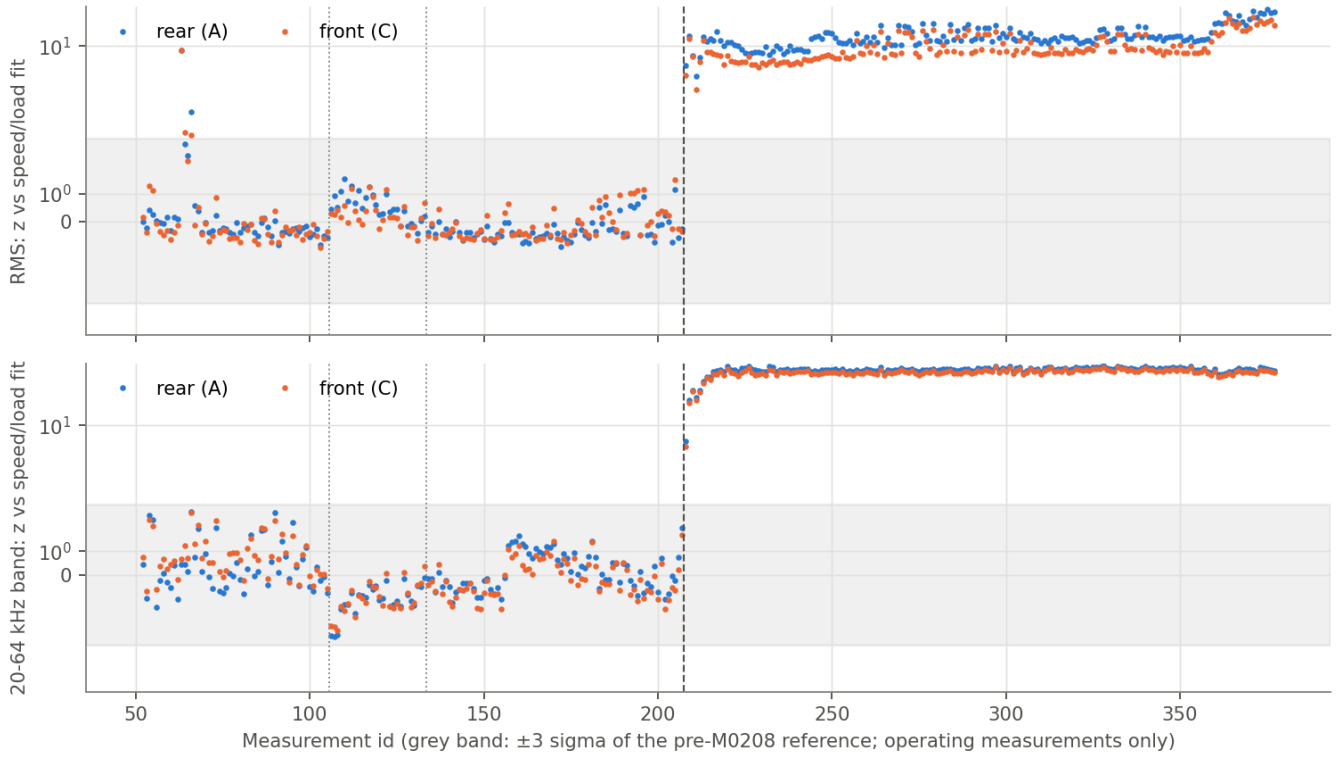
6. Looking for deterioration

Method

Vibration depends on speed and load, so raw levels cannot be compared across measurements. For each vibration feature I fitted, on the operating measurements before M0208 (156 measurements), ln(feature) = b0 + b1·ln(measured speed) + b2·static load + b3·dynamic peak load, and expressed every measurement as z, its distance from that fit in standard deviations. A change counts as sustained at the first measurement starting three in a row with z > 3. Temperature is left out of the fit on purpose and tested separately, because it rises with running time and possibly with damage.

Result: one abrupt change at M0208 (10:06:24)

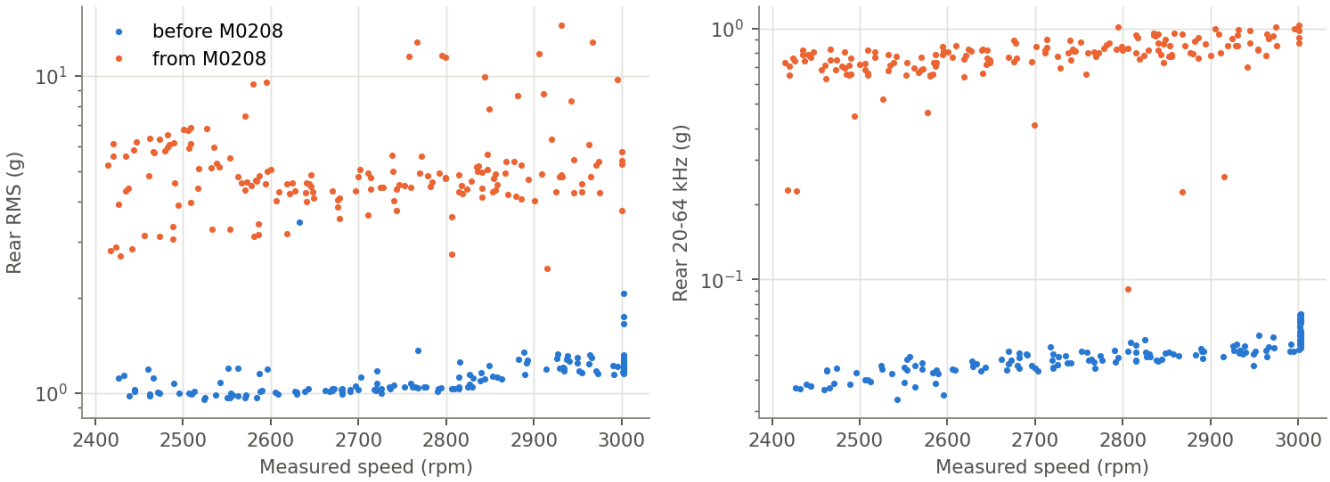
Feature (rear / front)	First sustained change	Median z afterwards	Exceedances before M0208
RMS	M0208 / M0208	11.5 / 9.0	2 / 3
Peak	M0208 / M0208	13.8 / 13.7	2 / 2
20–64 kHz band	M0208 / M0208	33.8 / 32.0	0 / 0
5–20 kHz band	M0208 / M0208	10.6 / 23.2	3 / 0
1–5 kHz band	M0208 / M0208	22.4 / 11.0	1 / 2
10 Hz–1 kHz band	M0208 / M0208	10.9 / 12.8	5 / 4
Kurtosis	M0208 / M0212	1.4 / 1.7	2 / 5



RMS and 20–64 kHz band after removing speed and load. Grey band: ±3 standard deviations of the reference before M0208. Dotted lines: the two stops.

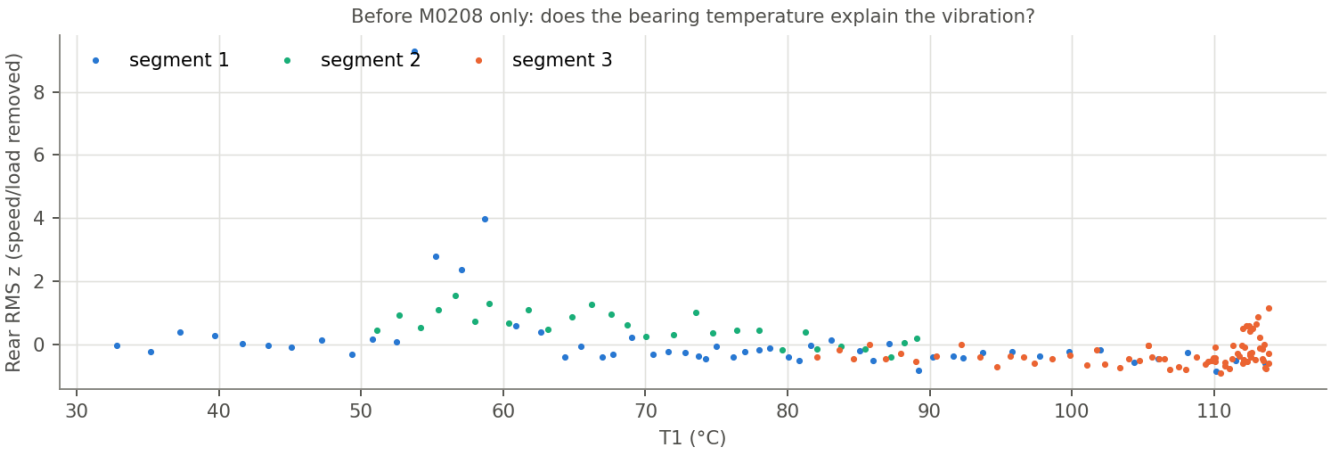
- **Abrupt:** the 20–64 kHz band goes from about 0.05 g to 0.09 g at M0208, 0.22 g at M0209 and about 0.8 g from M0216; RMS goes from about 1 g to 4–5 g within two measurements. Both accelerometers change together.
- **Not the operating conditions:** the 30 measurements before and after M0208 average 2778 vs 2699 rpm, 3528 vs 3573 N static, 928 vs 892 N dynamic, T1 112.8 vs 111.4 °C.
- **Robust:** with the reference ending at M0120, M0150, M0180 or M0200 instead, every feature still finds M0208 first, and the 20–64 kHz band never exceeds 3σ before it.

- **No warning beforehand:** no upward drift in the 50 operating minutes before M0208 (20–64 kHz: -0.01σ per 100 measurements).
- **Final escalation:** after about 45 minutes at a raised but stable level, vibration rises again from M0359 (10:53), kurtosis stays above 3.5 from M0367 and reaches about 6, and T1 jumps from 113 to 122.6 °C. Over the last 20 measurements RMS is 8.5× (rear) and 13.2× (front) its early level. The recording stops at 10:59:32 with no recorded reason.



Rear vibration against measured speed: from M0208 (orange) every measurement sits far above the earlier ones (blue) at every speed, so the change is not a speed effect.

Warm-up, restarts and temperature



Before M0208 only: rear RMS with speed and load removed, against bearing temperature T1, by segment.

- During the warm-up, vibration stays flat (+5% over 10 minutes) while T1 rises from 21 to 32 °C.
- After the cold restart (segment 2), RMS is about +0.5σ and the 20–64 kHz band about −1.6σ for a few minutes: a real but small restart effect, inside the normal band.
- Temperature is mostly a record of running history: it climbs after every start and levels off near 110 °C, and it does *not* rise at M0208; it rises only during the final escalation. Using it as a deterioration input without modelling warm-up would confuse "recently started" with "healthy".
- Measured RPM is set at random by the test bench: an operating condition used to normalise vibration, not a sign of condition.
- Short spikes at M0063–M0066 (segment 1) do not persist and the high-frequency band does not react; flagged, not removed.



Rear-accelerometer band levels, and kurtosis and crest factor of both accelerometers, by measurement number (stops collapsed).

7. Can B01 support deterioration labels and a health index?

Deterioration detection: partly

- **For:** a sharp, repeatable change on both sensors that speed, load and temperature do not explain.
- **Against:** no record of the bearing's actual condition (no inspection, and the change happens after the log ends); the change point comes from the same features a model would learn, so using it as a label would be circular; one bearing and one event, so nothing to validate on; no gradual onset, so no early-warning example; undocumented interventions between segments.
- **Honest use:** one unsupervised change-detection test case, with "normal" defined as the early operation.

0–100% Motor Health Index: not supported

- There is nothing to calibrate a percentage against: no wear or damage measurements and no documented failure.
- The run shows three regimes, not a continuum: normal (M0052–M0207), raised and stable (about M0208–M0358), and escalating (M0359–M0377). Turning these into percentages would be invented.
- What is supported: a distance-from-normal score (like the z-values here), shown as normal / abnormal / strongly abnormal. It is not remaining life or strength.

> **Hardware note.** The most sensitive feature here, the 20–64 kHz band, needs a sample rate of at least 128 kHz. At the 42 kHz used by the Ottawa pipeline it is lost; the 5–20 kHz band and RMS would still have shown this change, with less margin. Choose our sensor's sample rate and bandwidth with this in mind.

8. Do we need more Paderborn experiments before training?

Yes. In order of value:

1. **More run-to-failure experiments** of the same series (B02, B03, …) under the same random protocol, to see whether the change is always abrupt, how it varies between bearings, and to hold some bearings out for testing.
2. **Documentation from the Paderborn team** for B01: what happened at 09:13–09:32 and 09:40–09:43, whether segments 2 and 3 are the same bearing, why it stopped at 10:59, and the post-test inspection (damage type and size).
3. **Healthy-only runs** under the same random speeds and loads, long enough to cover warm-up, restarts and the temperature plateau, to measure the false-alarm rate.
4. **For a percentage scale:** runs with measured intermediate damage levels (inspections between stages or seeded defects of increasing size).
5. **For our own recordings:** sample vibration at 128 kHz (or at least keep the 5–20 kHz band), synchronised with temperature and measured RPM.

9. Files

Path (in `model/`)	What it is
`b01_preprocess.py`	reads all files, computes features, merges, runs the integrity checks (about 15 s)
`b01_analysis.py`	speed/load normalisation, change detection, temperature tests, figures
`results/B01/B01_processed.csv`	377 measurements × 47 columns, in time order
`results/B01/checks.json`, `analysis.json`	every check and analysis number
`results/B01/B01_analysis_report.md`	the detailed written report
`results/B01/figures/`	the five graphs in this report
`newdata/B01/`	the raw dataset (read only; now ignored by git)